

GSoC '17 Project Proposal

General Improvements and Bug Fixes for Tiled

BASIC INFORMATION

Personal Info

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University Info

- **University:** Indian Institute of Technology, Roorkee
- **Major:** Computer Science and Engineering
- **Current Year:** 2nd year (2019 expected graduation)
- **Degree:** Bachelor of Technology (B.Tech)

CODING SKILLS

Programming Languages and Frameworks

- Fluent in C/C++, Python, PHP and Javascript.
- Worked on Unreal Engine in C++, and web games in HTML5/Javascript.
- Experience in Laravel (PHP) and Node.js
- Also worked with Jekyll.
- Moderate knowledge of C# (Used in Unity3D), Ruby.

Development Environment

- Ubuntu 16.04 LTS or Windows 10 (I work on former most of the time)
- Sublime Text as primary editor. Fully customised for efficient use.
- Vim for small edits.

Version Control

- Strong experience in Git
- Also worked using SVN.

PRE-GSOC INVOLVEMENTS

Contribution to Tiled (Code Samples)

I've made a few pull requests to Tiled, and got most of them merged. I've mostly worked towards custom commands for user, but I've reviewed some small pull requests as well to get an idea of the codebase and working of Tiled. Here is a list of my contribution towards Tiled:

- [#1457](#) (merged): Fixes [#1127](#) (Adds an eye icon instead of checkbox)
- [#1460](#): Fixes [#943](#) (Adds option to run first enabled command from menu)
- [#1463](#) (merged): Options to run commands from menu as well as edit them
- [#1480](#) (merged): Same QAction to all QMenu
- [#1482](#) (merged): Custom commands for shortcuts
- [#1485](#) (merged): Added executable files in tests directory to gitignore
- [#1488](#) (merged): Added an executable picker - Fixes [#942](#)

PROJECT INTRODUCTION

Tiled is a map editor. A good user experience as well as interface is one of the key components for its users. A lot of small improvements can be done to Tiled. A lot of features have been mentioned in the issue tracker but have remained stagnant since then. I plan to add some of these features by the end of GSoC period and keep fixing bugs reported during the GSoC period.

DETAILED DESCRIPTION

Supporting Infinite Maps ([#260](#)):

- When creating a new map, we can initialize it with a fixed size (width x height). Even though we can resize it when required, it will be a very useful feature if the map size gets resized automatically.
- In other words, the canvas is supposed to be infinite and the user can paint anywhere
- This can be done by storing chunks of tiles of fixed size instead of storing each tile and then storing these chunks individually.
- The size of chunks will have to be experimented with to find the optimal value.
- A similar implementation: <https://github.com/Squarific/TiledCanvas>

Layer Locking ([#734](#)):

- Currently there is an option to make a particular layer visible or invisible. This helps when a user wants to paint on a particular layer without getting distracted by the other layers.
- Many editors like photoshop, GIMP, etc also allow layer locking so that a user can work on a particular layer without worrying about painting on other layers.

Autocrop ([#642](#)):

- This can be a useful feature, when the user isn't sure in the beginning as to what exactly the size of the map would be. The user can draw according to his/her needs and then use the autocrop feature to resize the map to their required needs.

Rotating Buttons in the Toolbar (#1084):

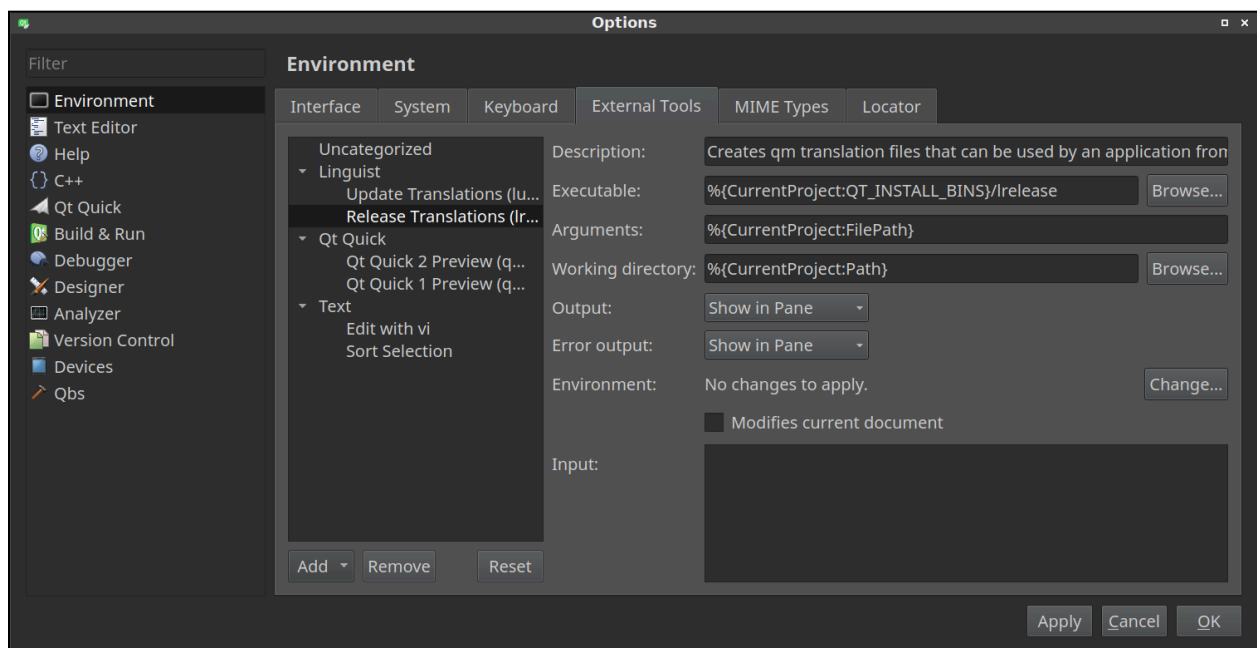
- Currently, only keyboard shortcuts for rotating the map as well as flipping it horizontally/vertically exist. A user would have no idea of these unless he goes through the shortcuts once.
- A simple solution is to add Tool Buttons into the Tool Bar for these.
- A slightly complex solution involves using tool buttons as used in GIMP. This would be part of my **wishlist** and I'll try to complete this after I've completed all other mentioned tasks.

Move Selection (#650):

- This is an important feature and it lacks in Tiled.
- The current feature behaves very weirdly, as mentioned in the issue ticket.

Custom Commands (#941, #208):

- Tiled allows users to add custom commands for themselves. These commands would then be called from terminal.
- The goal would be to make it similar to the Qt Creator External Tools.



Qt Creator External Tools Dialog

- I would also refer the Qt Creator's source for this.

Polygon Editing Tools (#1487)

- The main code resides in `editpolygontool.cpp`, but it hasn't been appreciated very much and there are a lot of improvements that can be made to this.
- A lot of changes have been suggested so far ([#117](#), [#145](#), [#404](#), [#762](#) and [#882](#)) and all these suggest that we need to refactor the whole logic behind this tool and start from scratch.
- The current implementation can be taken for reference while making the new tool with all the features implemented.

PROPOSED TIMELINE

May 5 - May 30 (Community Bonding)

- Familiarising myself with the mentors and the codebase. I will dive deep into the code and learn all the ins and outs of the code with help of mentors.
- Tiled 1.0 is supposed to be released in April/May, so this would be the time that a lot of bugs might be reported. I'll try to fix some of them to get a fair idea of working of Tiled.
- Complete the Autocrop feature.

May 30 - June 5

- Start working on Custom Commands.
- Go through the code of Qt Creator.
- Close #941, #208

June 6 - June 12

- Add finishing changes and complete custom commands task.

June 12 - June 18

- Start working on Rotating Toolbar task.

June 19 - June 25

- Start working on the wishlist part of the Rotating Toolbar task.
- Buffer time to finish remaining tasks till this week.

June 26 - July 2 (Phase 1 Evaluation)

- Make a working prototype of move selection feature.
- Consult the mentors regarding this prototype and make necessary changes.
- Get this feature merged by the end of this week.

July 3 - July 9

- Start working on layer locking feature.
- Since this is not a big task, I expect to finish this and get this merged by the end of this week.

July 10 - July 16

- Go through the code of [TiledPaint](#), to see the working of infinite canvas.
- Start working on initial prototype.

July 17 - July 23

- Complete the infinite canvas task.

July 24 - July 30 (Phase 2 Evaluation)

- Start working on Polygon editing tools.
- Go through the current code and come up with the flow and working of the tool.

July 31 - August 6

- Get a working prototype ready for polygon editing.

August 7 - August 13

- Complete the task for polygon editing tools

August 14 - August 21

- Buffer period for incomplete work or fix bugs reported during the GSoC period.
- Completion of tasks mentioned in the wishlist.
- Prepare for final evaluation.

ABOUT ME

I am a 19 year old, second year student currently enrolled in Computer Science and Engineering (IV Year Course) at IIT Roorkee. I developed a passion for programming and game development in my school days. I am regularly contributing to OpenSource community since my freshman year

I am an active member of [SDSLabs](#) at IIT Roorkee, a bunch of passionate enthusiasts trying to foster software development culture in the campus. Most of our work is done according to a strict timeline with a lot of focus on test driven development and following best practices to structure code as per the design principles for the task.

MOTIVATION

My motivation for GSoC this year is getting started with the open source community. GSoC is a great program for introducing organizations with prospective contributors and when I saw this project, I thought this is something that I can do. I have worked previously on various projects but never in a big project with a real world impact. This prospect is very exciting to me.

I chose this particular organisation because it is very well aligned with my interests and also correlated to what I have worked on in the past. Other than that, this project provides a good opportunity to apply my learnings on a practical scale. This is my field of interest and therefore the natural inclination to this project.

ACKNOWLEDGEMENTS

Is there anything important missing that we should know?

- No.

Do you have any other commitments during the main GSoC time period?

- No.

Do you have exams or classes that overlap with this period?

- My classes for next semester will begin from 17th July so after this period, my time commitment will be reduced. I will still manage to work around 5 hours a day on weekdays and about 8-10 hours on weekends.

Do you plan to have any other jobs or internships during this period?

- No.

Do you have any other short term commitments during this period? (e.g family wedding, conference, volunteer projects, planned vacation days)

- I will be travelling from 7th May to 13th May, which is way before the Coding Phase.